Bounded Continuous Regression Models

Continuous Responses with Zero and One Inflation

Jorge Luis Bazán, University of São Paulo

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1. Introduction: Motivation and Scope

- Bounded continuous responses have support $0 < Y < 1$.
- Examples include proportions, rates, indexes, and transformed mortality measures.
- Gaussian regression can produce fitted values outside the support.
- Beta regression is the usual benchmark for rates and proportions (Ferrari and Cribari-Neto 2004).
- JSBt regression builds flexible bounded distributions from real-line baselines (Rodrigues et al. 2020; Piccirilli, De Bastiani, and Jorge Luis Bazán 2026).
- The fitted models use the open interval; exact zeros and ones require an inflated extension (Galvis et al. 2014).

1.1 Related Regression Models

- Mixed and mixture beta regression: Verkuilen and Smithson (2012).
- Bayesian mixed beta regression, with mean and precision predictors: Figueroa-Zúñiga et al. (2013).
- Augmented beta regression for zeros and ones: Galvis et al. (2014).
- Simplex mixed-effects regression: Qiu et al. (2008).
- Earlier JSBt mixed regression: the master's dissertation of Piccirilli (2021).

1.2 Examples of Bounded Continuous Data

Bounded continuous regression can be fitted when Y is measured on a finite interval:

- mortality rates and disease proportions;
- poverty, inequality, and social indexes;
- percentages transformed to the unit interval;
- clinical measurements expressed as rates;
- proportions observed repeatedly in longitudinal studies.

These data often need models that represent asymmetry, heavy tails, and high concentration near zero or one.

2. Bounded Continuous Distributions: Data Structure

For the unit interval,

$$0 < Y < 1.$$

The response cannot move outside the interval:

- fitted values must remain in the support;
- the density may be asymmetric;
- tail behavior near zero and one may be scientifically relevant;
- changes in location, scale, and shape may all matter.

2.1 JSBt Construction

Let W be a real-line random variable with density $g(w;\theta)$ and distribution function $G(w;\theta)$. Define

$$Q_H(y) = \log\left(\frac{y}{1-y}\right), \qquad 0 < y < 1.$$

The bounded response is

$$Y = \frac{\exp(W)}{1+\exp(W)}.$$

The induced density on $(0,1)$ is

$$f(y;\theta) = g\left\{\log\left(\frac{y}{1-y}\right);\theta\right\}\frac{1}{y(1-y)}.$$

The distribution function is

$$F(y;\theta) = G\left\{\log\left(\frac{y}{1-y}\right);\theta\right\}.$$

Source: Rodrigues et al. (2020).

2.2 JSBt Model Families

Two-parameter JSBt distributions:

- Logistic Gumbel, LGU;
- Logistic Reverse Gumbel, LRG.

Three-parameter JSBt distributions (Piccirilli, De Bastiani, and Jorge Luis Bazán 2026; Piccirilli, De Bastiani, and Jorge L. Bazán 2026):

- Logistic Exponential Gaussian, LEG;
- Logistic Power Exponential, LPE;
- Logistic Skew-Normal Type I, LSN1;
- Logistic Skew-Normal Type II, LSN2.

Additional JSBt cases from Rodrigues et al. (2020):

- Logistic-logistic, LLo;
- Logistic-normal, LN;
- Logistic-t, LT.

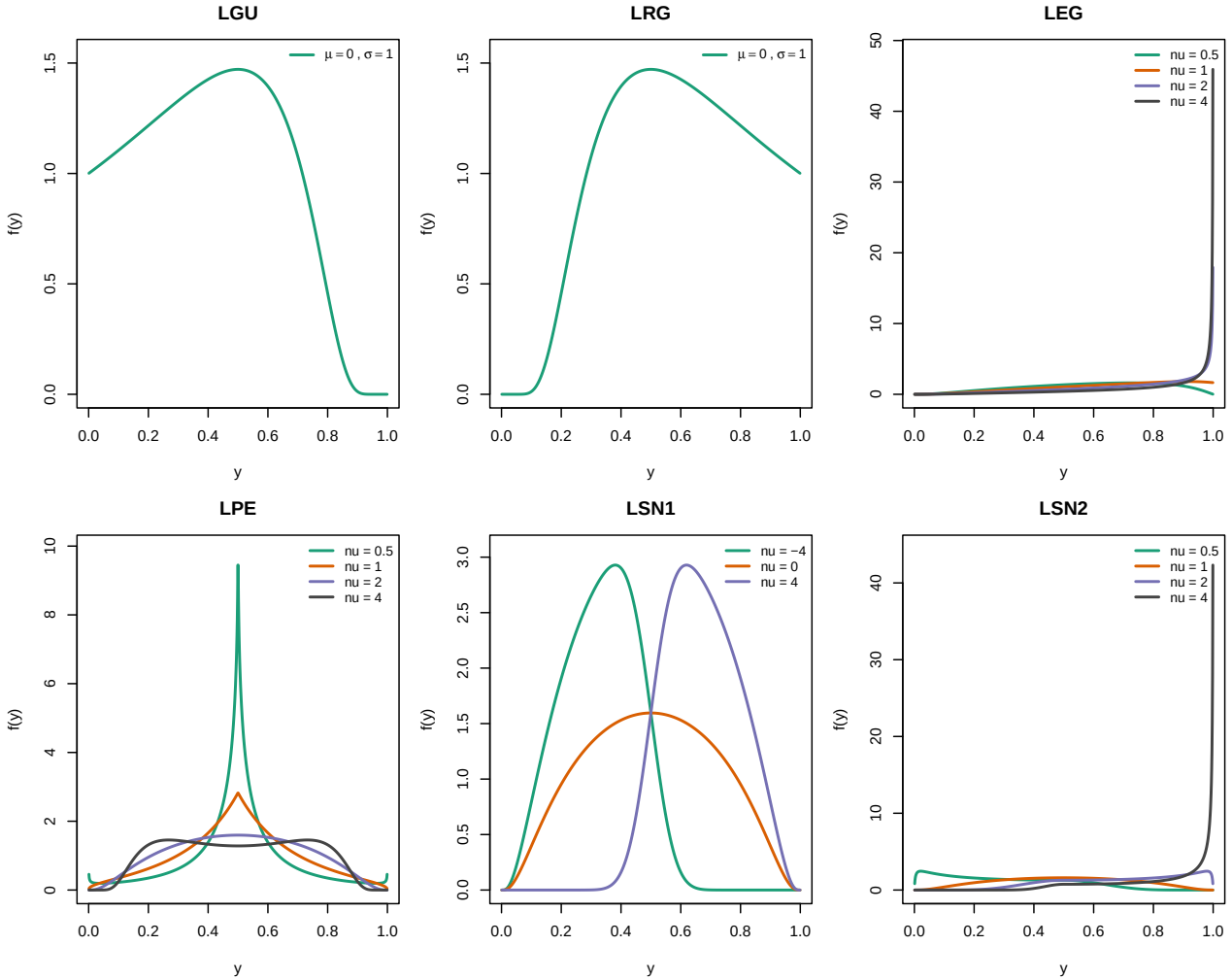
2.3 Parameter Interpretation

$$\theta = (\mu, \sigma) \quad \text{or} \quad \theta = (\mu, \sigma, \nu).$$

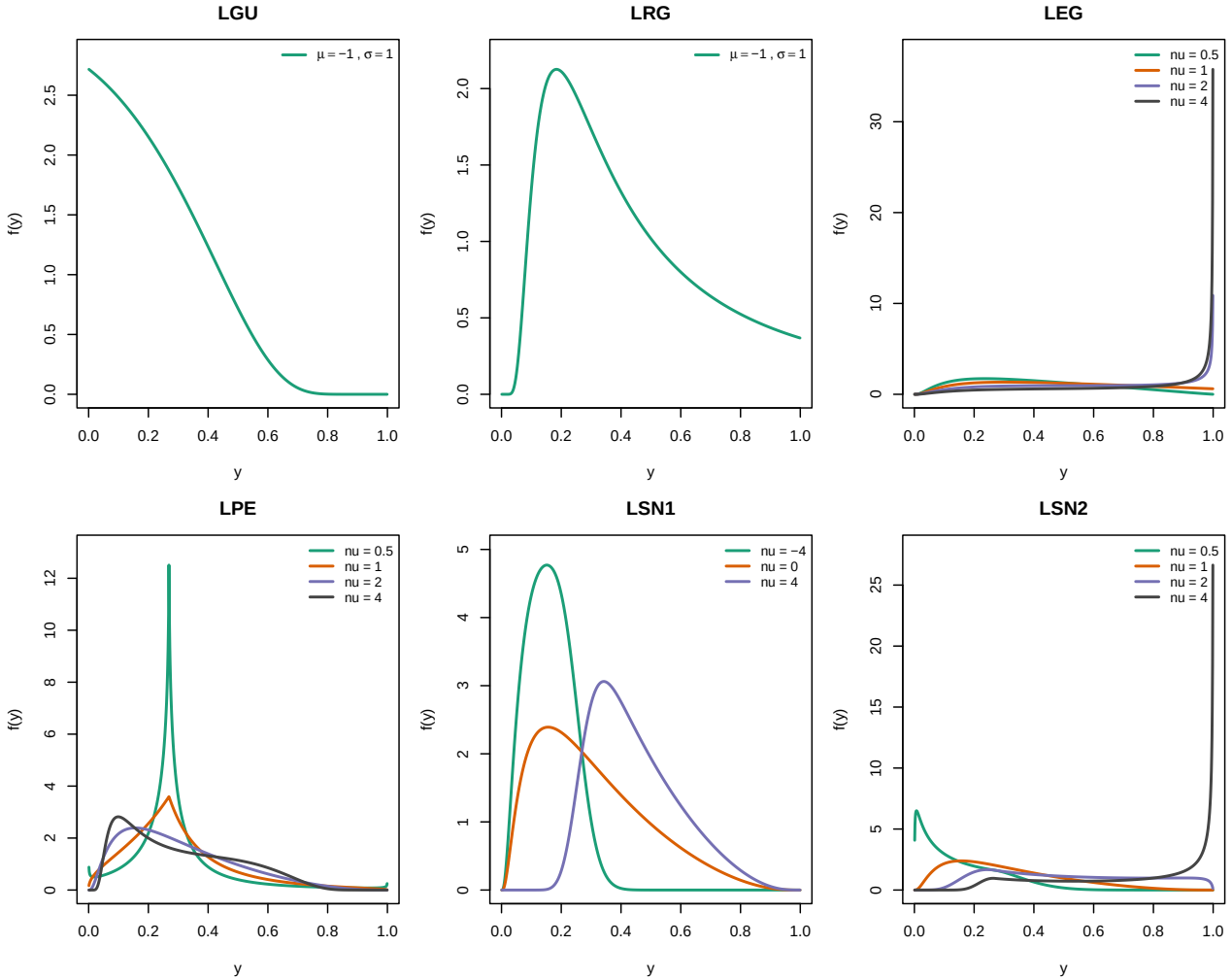
- μ is a location parameter on the logit scale.
- $\sigma > 0$ controls scale or dispersion on the logit scale.
- ν controls shape when the baseline family has a shape parameter.
- μ is not generally equal to $\mathbb{E}(Y)$.
- A useful response-scale location index is

$$\omega = \frac{\exp(\mu)}{1 + \exp(\mu)}.$$

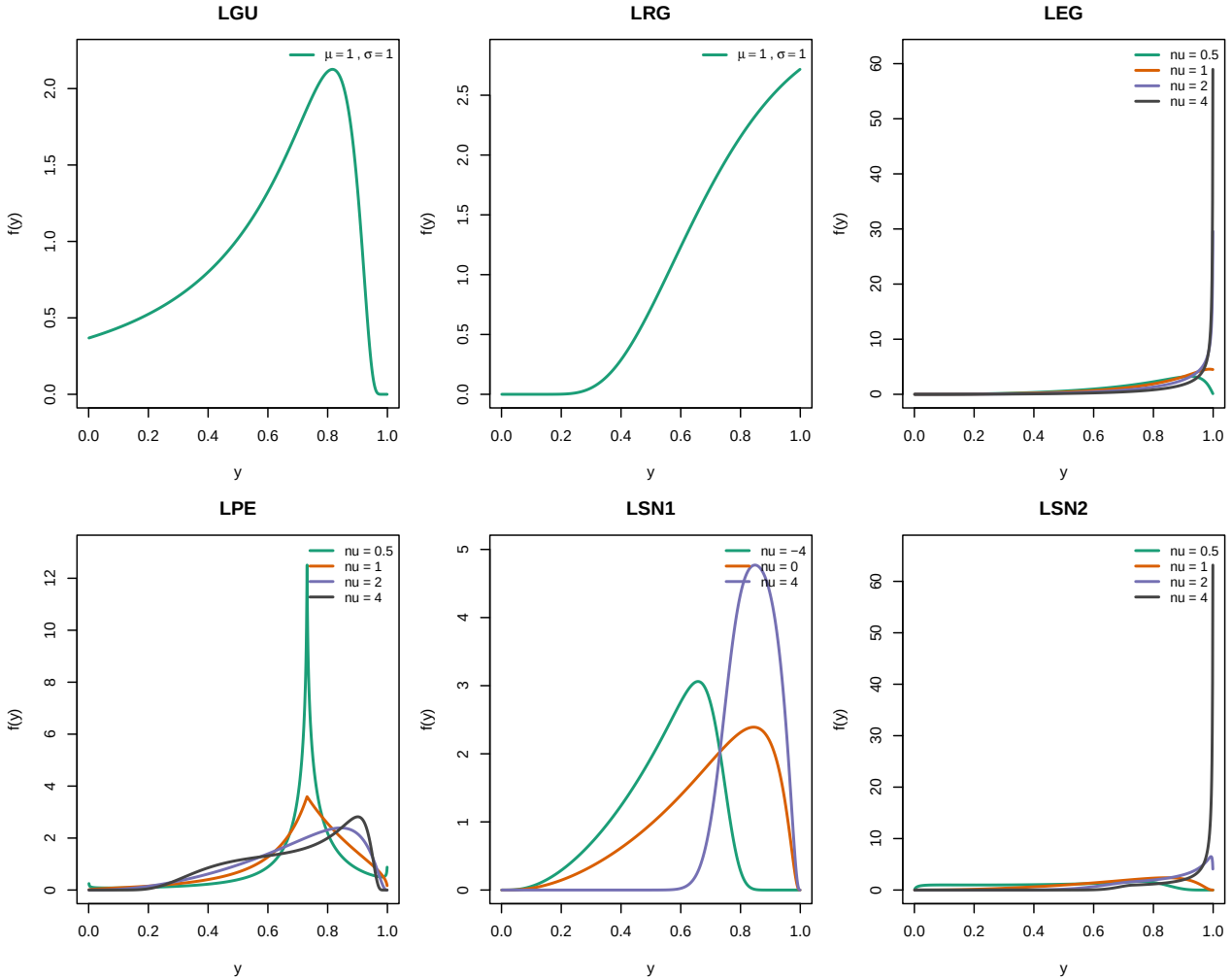
2.4 Density Shapes for $\mu = 0$



2.5 Effect of Location: $\mu = -1$



2.6 Effect of Location: $\mu = 1$



2.7 Mean on the Response Scale

Reference values from numerical integration:

Distribution	Reference	$\mathbb{E}(Y)$, $\mu = -1$	$\mathbb{E}(Y)$, $\mu = 0$	$\mathbb{E}(Y)$, $\mu = 1$
LGU	$\sigma = 1$	0.2283	0.4037	0.5964
LRG	$\sigma = 1$	0.4036	0.5963	0.7717
LEG	$\sigma = 1, \nu = 1$	0.4889	0.6708	0.8190
LPE	$\sigma = 1, \nu = 1$	0.2987	0.5000	0.7013
LSN1	$\sigma = 1, \nu = 0$	0.3033	0.5000	0.6967
LSN2	$\sigma = 1, \nu = 1$	0.3033	0.5000	0.6967

The location parameter is defined before the bounded transformation is applied.

2.8 Identification and Monotonicity

Identification follows from the one-to-one logit transformation:

$$Y = s(W) = \frac{\exp(W)}{1 + \exp(W)}, \qquad W = Q_H(Y).$$

If two bounded densities are equal on $(0, 1)$, their baseline densities are equal. Thus, an identifiable baseline induces an identifiable JSBt family.

For fixed σ and ν,

$$\Pr(Y > y) = \Pr\{W > Q_H(y)\}$$

is nondecreasing in μ.

Density curves can cross, so monotonicity in μ is stochastic rather than pointwise.

2.9 Skewness and Kurtosis

For $Y = s(W)$,

$$m_r = \mathbb{E}(Y^r) = \int_{-\infty}^{\infty} \{s(w)\}^r g(w;\theta)\, dw, \qquad r = 1, 2, 3, 4.$$

$$\text{Skew}(Y) = \frac{\mathbb{E}\{Y - \mathbb{E}(Y)\}^3}{\{\text{Var}(Y)\}^{3/2}},$$

$$\text{Kurt}(Y) = \frac{\mathbb{E}\{Y - \mathbb{E}(Y)\}^4}{\{\text{Var}(Y)\}^2}.$$

These quantities are generally evaluated numerically.

2.10 Shape Parameters

- LGU and LRG provide complementary asymmetric cases.
- LEG can affect boundary concentration and tail weight through $\nu > 0$.
- LPE uses $\nu > 0$ to control tail thickness and peakedness.
- LSN1 uses $\nu \in \mathbb{R}$ as a skewness parameter.
- LSN2 uses $\nu > 0$ as a two-piece skewness parameter.
- Shape modeling helps when two distributions have similar location but different tail or boundary behavior.

2.11 Additional Baselines and Parameterizations

- Logistic-logistic is the L-Logistic model, derived from the logistic baseline (Tadikamalla and Johnson 1982; Paz et al. 2019).
- Logistic-normal is the classical Johnson S_B model (Johnson 1949); symmetric-baseline extensions are studied by Lemonte and Bazán (2016).
- Logistic-t adds a degrees-of-freedom parameter ν; smaller values give heavier baseline tails.

For these symmetric baselines, the response median is

$$m = \text{logit}^{-1}(\mu).$$

For the L-Logistic parameterization of Paz et al. (2019),

$$b = 1/\sigma.$$

Thus larger b corresponds to smaller baseline scale σ.

3. Regression Development

- Rodrigues et al. (2020) introduce covariates through the baseline location, with constant scale and shape in their poverty-data application.
- Piccirilli (2021) presents an earlier treatment of JSBt mixed regression.
- Piccirilli, De Bastiani, and Jorge Luis Bazán (2026) develop location, scale, and shape regression, with random effects on location for longitudinal data. The article is listed as forthcoming.
- Here we present independent-response regression; the practical examples use intercept-only scale and shape models.

3.1 Regression Models: General Structure

For independent observations $(y_i, \mathbf{x}_i)$, $i = 1, \ldots, n$,

$$0 < y_i < 1.$$

For a three-parameter JSBt model,

$$Y_i \sim \text{JSBt}(\mu_i, \sigma_i, \nu_i).$$

$$g_1(\mu_i) = \mathbf{x}_{1i}^\top \boldsymbol{\beta}, \qquad g_2(\sigma_i) = \mathbf{x}_{2i}^\top \boldsymbol{\gamma}, \qquad g_3(\nu_i) = \mathbf{x}_{3i}^\top \boldsymbol{\delta}.$$

3.2 Links and Interpretation

- Identity links are natural for real-line location parameters.
- Log links are natural for positive scale and positive shape parameters.
- LSN1 may use an identity link for ν because $\nu \in \mathbb{R}$.
- Covariates can enter location, scale, and shape parameters.
- When interpretation on $(0, 1)$ is needed, transform fitted μ_i with

$$\widehat{\omega}_i = \frac{\exp(\widehat{\mu}_i)}{1 + \exp(\widehat{\mu}_i)}.$$

3.3 Regression Log-Likelihood

Let

$$w_i = \log\left(\frac{y_i}{1-y_i}\right).$$

The generic log-likelihood is

$$\ell(\Theta) = \sum_{i=1}^{n} \log g\left\{w_i; \mu_i, \sigma_i, \nu_i\right\} - \sum_{i=1}^{n} \log\{y_i(1-y_i)\}.$$

The Jacobian term accounts for the change of variable from W to Y.

3.4 Model Comparison Criteria

For a fitted model with log-likelihood ℓ, p effective parameters, and sample size n:

$$\text{Deviance} = -2\ell, \qquad \text{AIC} = -2\ell + 2p,$$

$$\text{BIC} = -2\ell + p\log(n).$$

AIC and BIC balance fit against complexity. Compare models fitted to the same observations and response scale, alongside quantile residuals (Dunn and Smyth 1996).

4. Practical Estimation: GAMLSS and Reproducibility

The notebook uses GAMLSS (Rigby and Stasinopoulos 2005; Stasinopoulos and Rigby 2007):

- `BE` for the beta benchmark;
- `gamlss.dist::gen.Family()` to generate JSBt families (Stasinopoulos and Rigby 2023);
- `logitGU`, `logitRG`, `logitexGAUS`, `logitPE`, `logitSN1`, and `logitSN2`;
- `logitLO`, `logitNO`, and `logitTF` for additional JSBt baselines.

Each distributional parameter can have its own predictor (Rigby et al. 2019).

The slides reuse the analysis in:

`BoundedContinuosRegression.Rmd`.

4.1 Simulated Data for JSBt Regression

The same covariate values are used in both simulations:

$$n = 500, \qquad x_i \sim N(0, 1), \qquad \eta_i = -0.40 + 1.20 x_i.$$

Beta-generated case (GAMLSS `BE` parameterization):

$$Y_i \sim \text{Beta}(\mu_i, \sigma), \qquad \mu_i = \text{logit}^{-1}(\eta_i), \qquad \sigma = 0.20.$$

Logistic-LSN1-generated case:

$$Y_i \sim \text{logitSN1}(\mu_i, \sigma, \nu), \qquad \mu_i = \eta_i, \quad \sigma = 0.45, \quad \nu = 5.$$

In the beta case, μ_i is the response mean and σ is beta dispersion. In LSN1, μ_i is baseline location and σ is baseline scale.

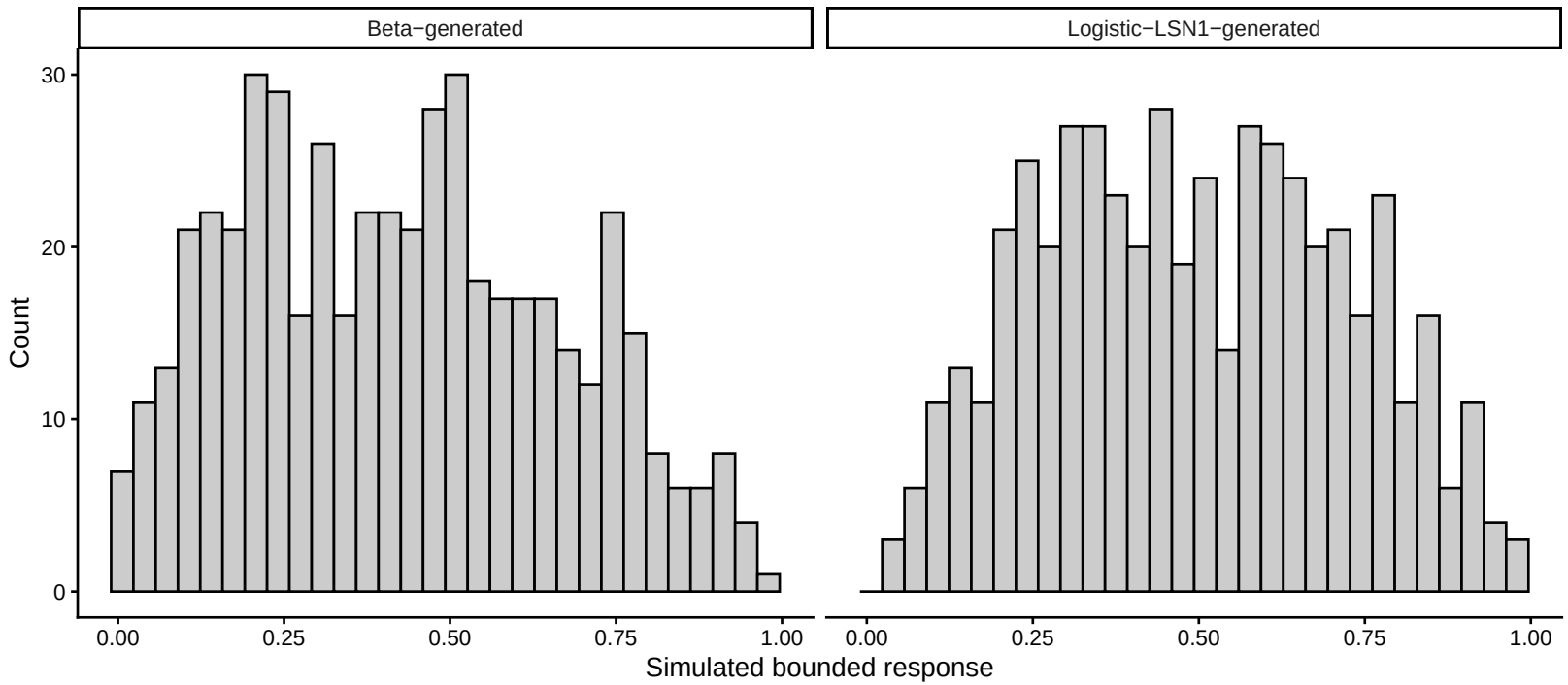
4.2 Simulated Response Summaries

Response summaries:

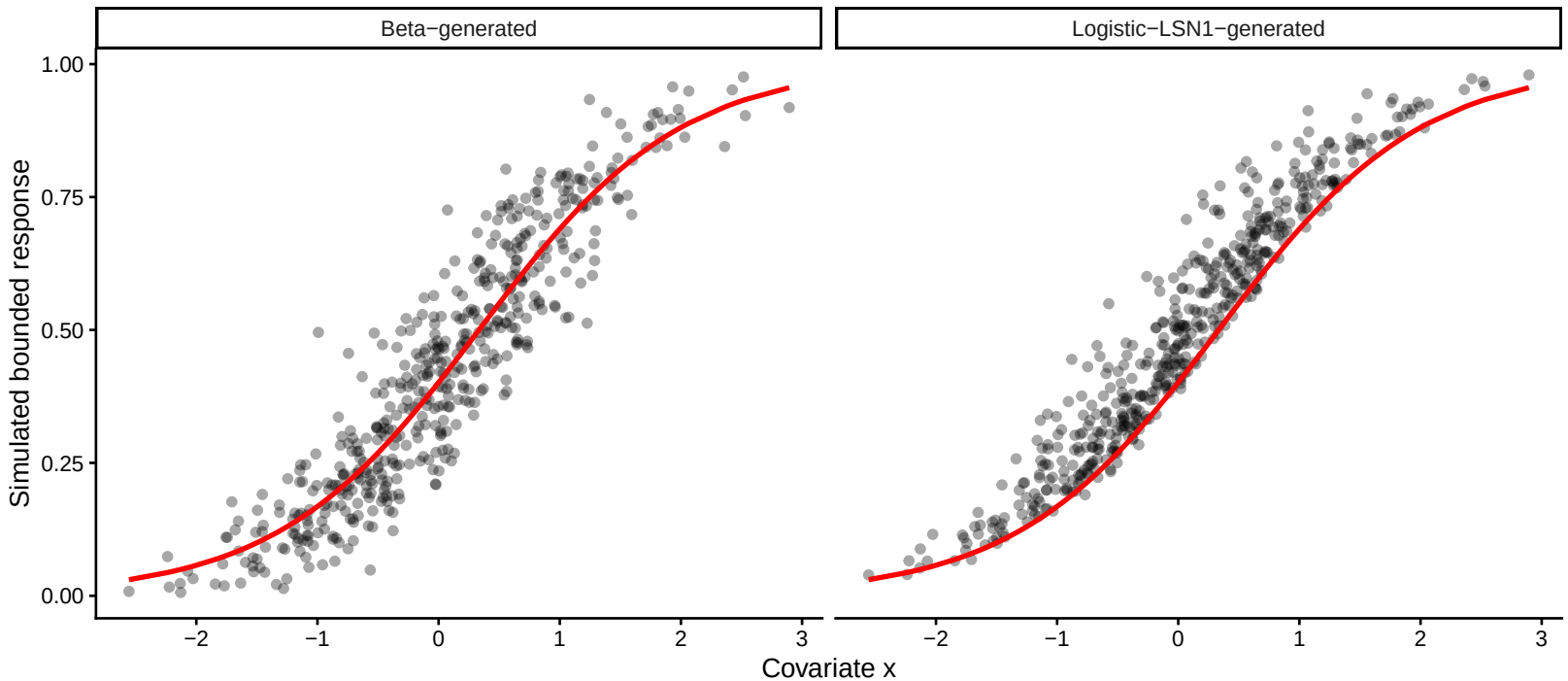
Scenario	Min	Median	Mean	Max
Beta-generated	0.006	0.418	0.426	0.976
Logistic-LSN1-generated	0.039	0.485	0.493	0.980

- The beta case uses a direct response-scale mean regression.
- The LSN1 case uses a real-line location regression and then transforms to $(0, 1)$.
- Both examples keep the remaining distributional parameters fixed.

Simulated Response Distributions



Simulated Responses and True Location



Red: conditional mean for beta; inverse-logit baseline location for LSN1.

4.3 Fitting Candidate Models

```
fit_regression_models <- function(dat) {
  list(
    Beta = gamlss(y ~ x, sigma.formula = ~ 1,
      data = dat, family = BE),
    LGU = gamlss(y ~ x, sigma.formula = ~ 1,
      data = dat, family = logitGU),
    LRG = gamlss(y ~ x, sigma.formula = ~ 1,
      data = dat, family = logitRG),
    LSN1 = gamlss(y ~ x, sigma.formula = ~ 1,
      nu.formula = ~ 1, data = dat, family = logitSN1)
  )
}
```

The full notebook also fits LLo, LN, LEG, LPE, LSN2, and LT. Scale and shape are estimated as constants; only location (or the beta mean) depends on x.

4.4 Model Comparison: Beta-Generated Data

Model	AIC	BIC
Beta	-1089.60	-1076.96
LLo	-1054.90	-1042.25
LT	-1054.19	-1037.33
LPE	-1049.53	-1032.68
LSN2	-1045.99	-1029.13
LN	-1043.26	-1030.62
LSN1	-1041.26	-1024.40
LEG	-1041.24	-1024.38
LGU	-984.00	-971.36
LRG	-888.52	-875.87

Minimum AIC: **Beta**. Minimum BIC: **Beta**.

Beta is the data-generating benchmark; selection is based on the realized sample.

4.5 Model Comparison: LSN1-Generated Data

Model	AIC	BIC
LSN1	-1568.09	-1551.24
LSN2	-1565.06	-1548.20
LRG	-1562.87	-1550.23
LEG	-1555.59	-1538.73
LLo	-1479.29	-1466.65
LT	-1477.43	-1460.57
LPE	-1472.25	-1455.39
LN	-1467.49	-1454.85
Beta	-1452.11	-1439.46
LGU	-1258.16	-1245.51

Minimum AIC: **LSN1**. Minimum BIC: **LSN1**.

LSN1 is the data-generating benchmark; selection is based on the realized sample.

4.6 Residual Diagnostics

For continuous responses, the quantile residual is (Dunn and Smyth 1996)

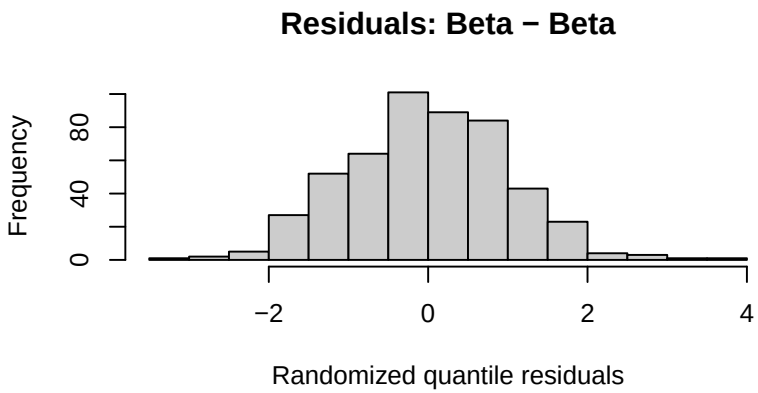
$$r_i = \Phi^{-1}\{F(y_i; \widehat{\theta}_i)\}.$$

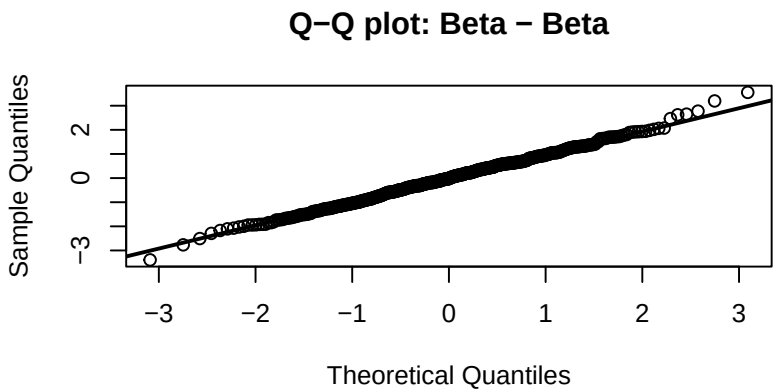
Randomization is needed when the fitted distribution has discrete masses.

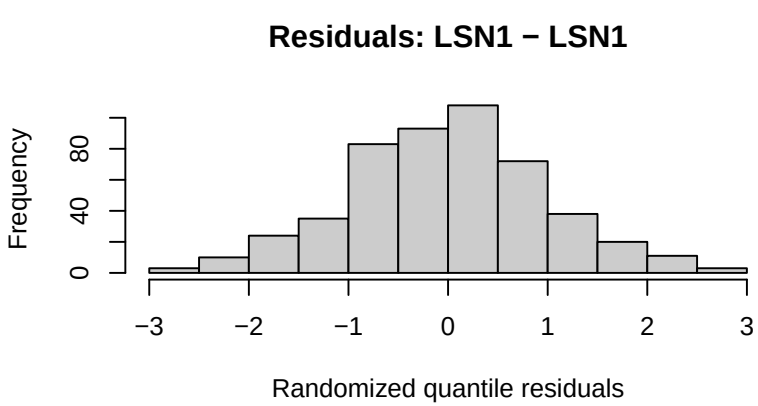
Scenario	Model	ShapiroP	Mean	SD
Beta	Beta	0.717	-0.004	1.000
LSN1	LSN1	0.558	-0.004	1.001

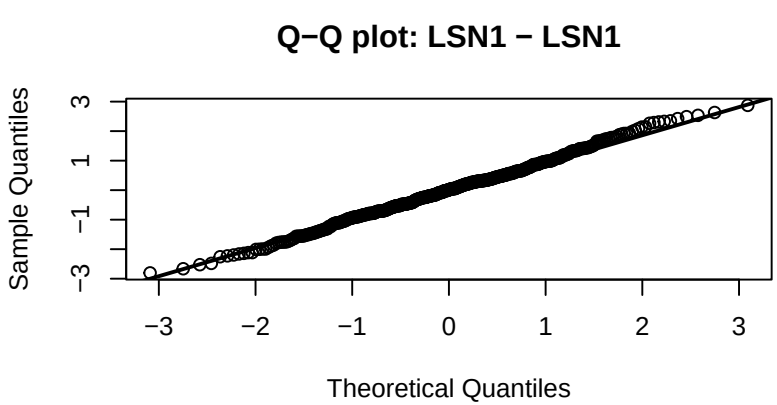
An adequate fit should yield approximately standard normal residuals. Read the Shapiro–Wilk p-values together with the histograms and Q–Q plots.

Residual Histograms and Q–Q Plots









4.7 Fitted Location on the Response Scale

For beta regression:

$$\widehat{\mu}_i$$

is already the fitted conditional mean on $(0, 1)$.

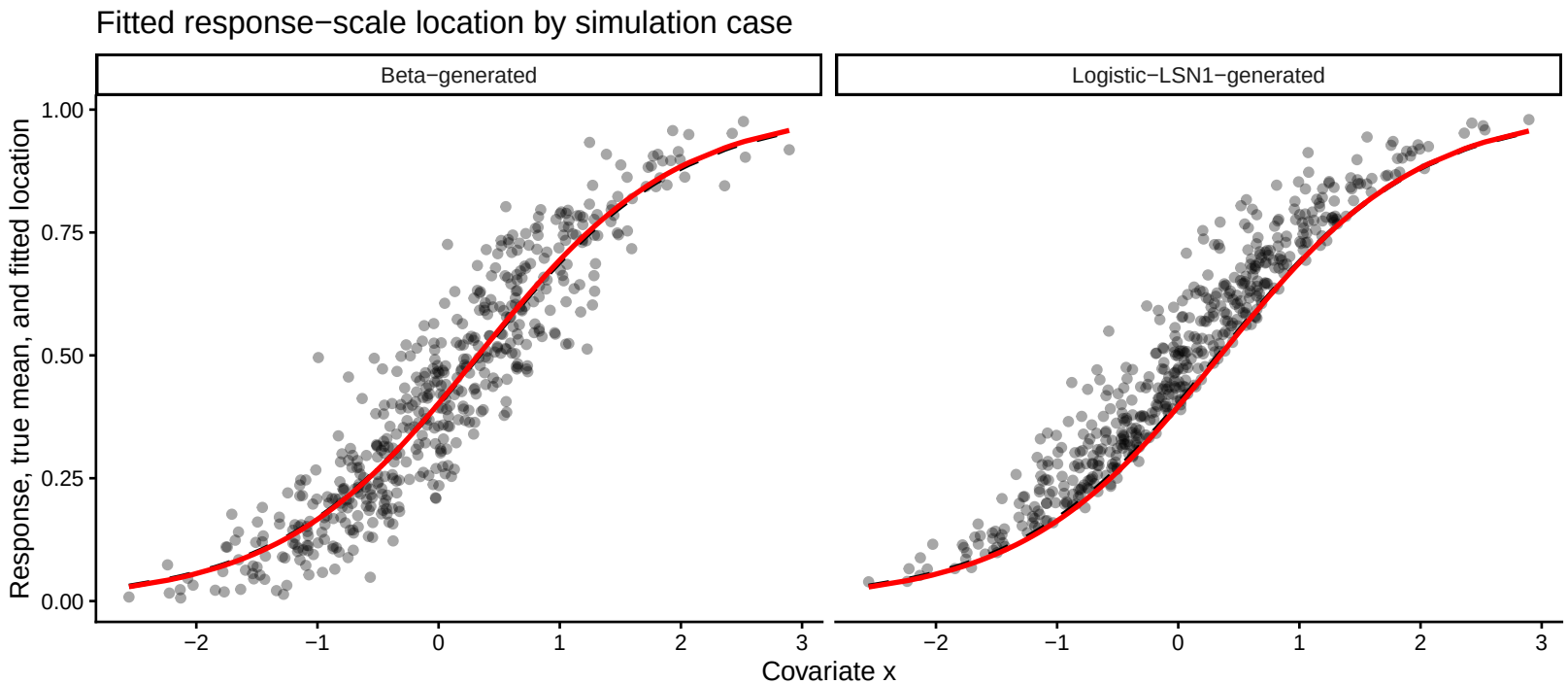
For JSBt regression:

$$\widehat{\omega}_i = \text{logit}^{-1}(\widehat{\mu}_i)$$

is a response-scale location index.

For asymmetric JSBt models, $\widehat{\omega}_i$ is not necessarily the conditional mean, but it is useful for interpreting how covariates move the center of the bounded distribution.

Fitted and True Response-Scale Location



Dashed: true mean (beta) or location index (LSN1). Red: fitted counterpart.

5. Exercises: Mortality and Logistic Models

1. Fit JSBt models to the second application (mortality) dataset in Piccirilli, De Bastiani, and Jorge Luis Bazán (2026); compare AIC, BIC, and residual diagnostics. Dataset.
2. Simulate Logistic-logistic data and compare GAMLSS with `llbayesireg` and `unitquantreg`, accounting for parameterization (Fonsêca and Paz 2019; Menezes and Mazucheli 2021; Mazucheli et al. 2022).

5.1 Exercises: Other Benchmarks and Bayesian Estimation

3. Use `GasolineYield` from `betareg` to compare beta and JSBt regression (Cribari-Neto and Zeileis 2010).
4. Use `sdac` from `simplexreg` to compare simplex and JSBt regression (Zhang et al. 2016).
5. Implement Bayesian estimation of the models introduced here.

6. Conclusions

- Bounded continuous models must respect the support $0 < Y < 1$.
- Beta regression is the benchmark when the mean-precision structure is adequate.
- JSBt regression generates flexible bounded models from real-line baseline distributions.
- Location, scale, and shape can be modeled through covariates in the GAMLSS framework.
- Model choice should combine AIC, BIC, quantile residuals, and substantive interpretation.
- `BoundedContinuosRegression.Rmd` reproduces the derivations, simulations, and figures.

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